SOFTWARE ENGINEERING AND DESIGN PRINCIPLES

(CS311L)

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AI CHATBOT FOR COLLEGE HELPDESK

Identify the classes. draw the class diagram.

1. Identified Classes

No.	Class	Purpose
1	User	General superclass for people interacting with the chatbot.
2	Student	Represents students, including authenticated students accessing personalized information.
3	FacultyStaff	Represents faculty/staff users accessing internal or administrative information.
4	Visitor	Represents guest users restricted to general information.
5	ChatSession	Represents a chatbot conversation/session and conversational state.
6	Query	Represents a user question, its text, intent and confidence.
7	KnowledgeBase	Stores categorized FAQ/question-answer entries.
8	Ticket	Represents an escalated helpdesk request and its status/category.
9	HelpdeskOperator	Handles escalated tickets and monitors conversations.
10	Admin	Provides privileged administration functions such as knowledge-base management and analytics.

2. Class Specifications

User

Attributes	Operations
userId : String name : String email : String role : String	askQuery() viewInformation()

Student

Attributes	Operations
rollNo : String	authenticate()

authenticated : Boolean	viewPersonalizedInfo()

FacultyStaff

Attributes	Operations
employeeId : String department : String	askAdminQuery() viewInternalInfo()

Visitor

Attributes	Operations
visitorId : String	viewGeneralInfo()

ChatSession

Attributes	Operations
sessionId : String startTime : DateTime status : String	sendMessage() maintainContext()

Query

Attributes	Operations
queryId : String text : String intent : String confidence : Float	classifyIntent() extractEntities()

KnowledgeBase

Attributes	Operations
entries : List categories : List	addEntry() updateEntry() deleteEntry() search()

Ticket

Attributes	Operations
ticketId : String category : String status : String createdAt : DateTime	create() updateStatus() route()

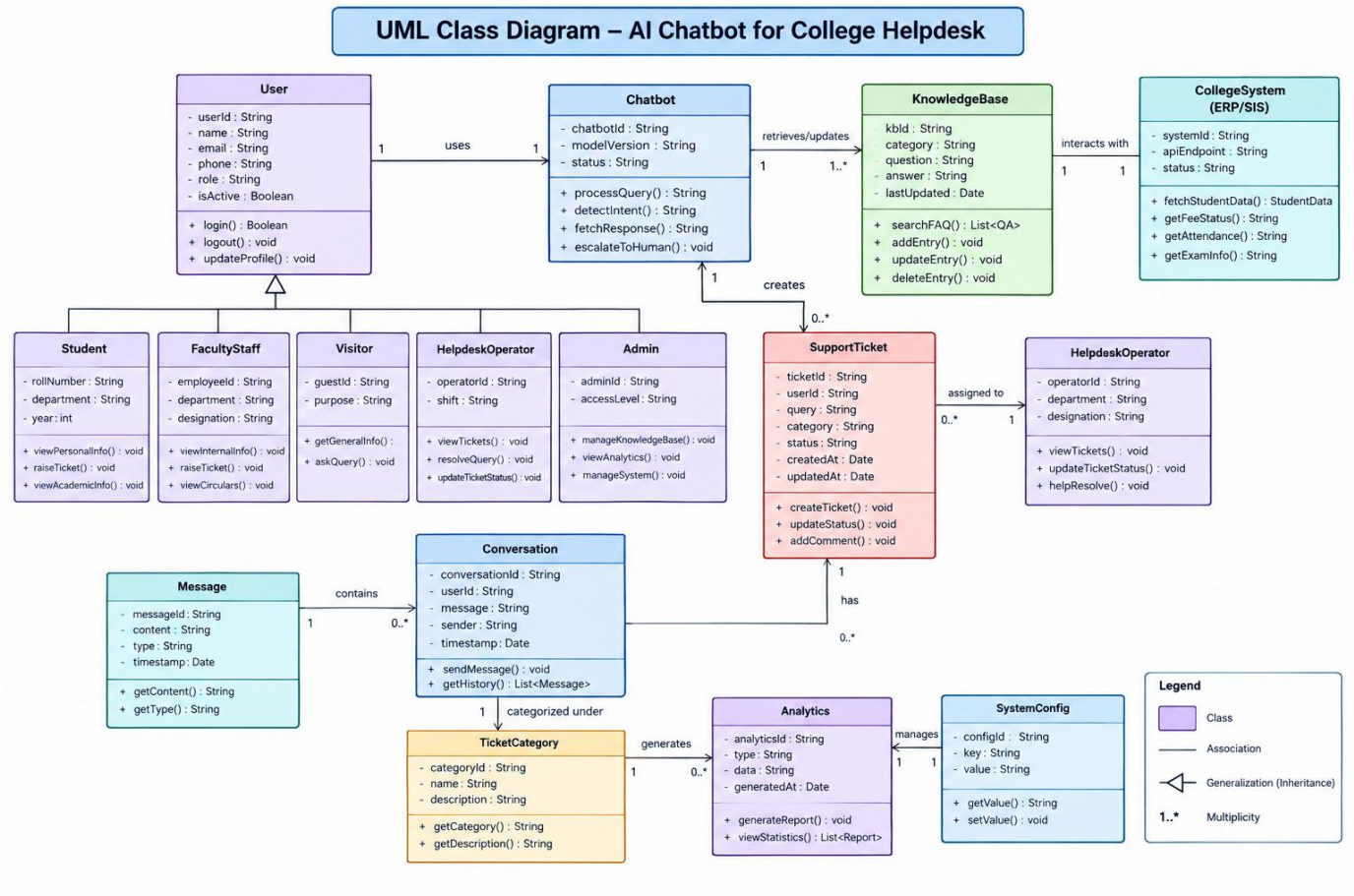
HelpdeskOperator

Attributes	Operations
operatorId : String department : String	resolveQuery() manageTicket() monitorChats()

Admin

Attributes	Operations
adminId : String privileges : String	manageKB() viewAnalytics()

3. UML Class Diagram



4. Relationships Between Classes

- **Student —|> User:** Student inherits common user properties and behaviour from User.
- **FacultyStaff —|> User:** Faculty/Staff is a specialized User type.
- **Visitor —|> User:** Visitor is a specialized User type.
- **Student/FacultyStaff/Visitor — ChatSession:** Users interact with the chatbot through conversation sessions.
- **ChatSession — Query:** A session contains and processes user queries.
- **Query — KnowledgeBase:** Queries can be matched with FAQ/knowledge-base entries.
- **Query — Ticket:** Unresolved or explicitly escalated queries can create tickets.
- **HelpdeskOperator — Ticket:** Operators handle and update escalated tickets.
- **Admin — KnowledgeBase:** Administrators manage knowledge-base entries.
- **Admin — HelpdeskOperator:** Admin represents higher-level administrative control over helpdesk functions.